

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Beat Optical and ImageNet Transfer Across ViT and CNN?

Anonymous WACV Applications Track submission

Paper ID *****

Abstract

001 Detecting vessels without an Automatic Identification
002 System correlate (dark vessels) in spaceborne syn-
003 thetic aperture radar (SAR) is operationally impor-
004 tant, yet human-verified labels are scarce. We study
005 how pretraining domain affects label efficiency on
006 xView3-SAR [9] under one fixed point-native detector.
007 Two size-matched tracks—ViT-B/16 (~86M param-
008 eters) and ConvNeXt-V2-Base (~89M)—each compare
009 random, generic ImageNet-1K, optical remote-sensing,
010 and SAR initialization at nested 10/25/50/100% scene-
011 level training fractions. Within each track, architec-
012 ture, input, detector, optimization, and the fixed run
013 seed (0) are held constant; only encoder initialization
014 changes. All six pretrained encoders are exact down-
015 loaded checkpoints, and this project performs no back-
016 bone pretraining. The controlled matrix contains 32
017 seed-0 fine-tunes; two external references are reported
018 separately. We report detection-F1 label-efficiency
019 curves as point estimates, near-shore F1, and—on
020 a disjoint, once-touched 50-scene verified set—dark-
021 vessel recall. Differences in pretraining objective and
022 history are stated explicitly when interpreting domain
023 and architecture gaps.

024 1. Introduction

025 Detecting ships without an Automatic Identification
026 System (AIS) correlate—dark vessels—in spaceborne
027 SAR is a critical maritime-monitoring problem with
028 scarce labels. xView3-SAR [9] provides Sentinel-1 im-
029 agery, but its human-verified dark-vessel examples are
030 limited to a small held-out set.

031 Pretrained backbones can help, but it is unclear
032 which source transfers best: generic natural images
033 (ImageNet-1K [4]), optical remote sensing (fMoW-
034 RGB [2] via SatDINO [11], or BigEarthNet-S2),

or SAR imagery (SAR-1M via SARMAE [8], or
BigEarthNet-S1)—and whether the answer is the same
for a ViT [5] and a ConvNeXt [12]. We compare ex-
act released checkpoints rather than pretraining our
own, avoiding a confound from locally chosen source-
training recipes [14]. The released checkpoints are not
all method-matched, so pretraining objective and his-
tory remain explicit interpretation limits rather than
hidden assumptions.

With the detector, schedule, and input fixed, we
ask: (Q1) how much labeled xView3 data does each
initialization save relative to random initialization to
reach a target overall detection F1; (Q2) does SAR-
domain pretraining beat both optical remote-sensing
and generic ImageNet transfer in the scarce-label
regime; and (Q3) is any such advantage architecture-
general, holding for both a ViT and a size-matched
CNN? We hypothesize that SAR initialization yields
the largest low-label gain over the random floor in both
tracks. Because the approved study uses seed 0 only,
its curves are descriptive point estimates, not variance
or significance estimates.

2. Approach

Design. The study has two architecture-matched
tracks—ViT-B/16 (~86M) and ConvNeXt-V2-Base
(~89M, deliberately size-matched)—each with four
arms that share the same head, optimizer, schedule,
and fixed seed 0; only the downloaded initialization
differs (Table 1). Initialization is the only variable
within a track and architecture the only variable across
matched roles. Comparing arm k to arm $k+4$ holds
the pretraining role fixed while changing architecture,
subject to the disclosed cross-track training-history dif-
ferences.

Initialization regimes. The four matched roles are
a random floor, generic natural-image transfer, opti-

Arm	Track	Backbone	Initialization	Params	On curve?	Role
1	ViT	ViT-B/16	random init	~86M	yes	ViT floor
2	ViT	ViT-B/16	SatDINO (fMoW-RGB, DINO)	~86M	yes	optical-domain transfer
3	ViT	ViT-B/16	SARMAE (SAR-1M, MAE)	~86M	yes	SAR-domain transfer
4	ViT	ViT-B/16	ImageNet-1K AugReg	~86M	yes	generic ImageNet transfer
5	CNN	ConvNeXt-V2-B	random init	~89M	yes	CNN floor
6	CNN	ConvNeXt-V2-B	BigEarthNet-S2 (optical RS)	~89M	yes	optical-domain transfer
7	CNN	ConvNeXt-V2-B	BigEarthNet-S1 (SAR RS)	~89M	yes	SAR-domain transfer
8	CNN	ConvNeXt-V2-B	ImageNet-1K FCMAE→FT	~89M	yes	generic ImageNet transfer
R2	ref	YOLO26	COCO-pretrained	external	no	detector reference
R3	ref	LocateAnything-3B	zero-shot	~3B	no	VLM reference

Table 1. Eight controlled arms in two size-matched tracks (~86M vs. ~89M), plus two external references reported off the curves. Optical anchors are SatDINO [11] (ViT) and BigEarthNet-S2 [3] (CNN); SAR anchors are SARMAE [8] (ViT) and BigEarthNet-S1 [3] (CNN). Both ImageNet encoders are downloaded; Arm 4 uses supervised AugReg while Arm 8 uses FCMAE followed by supervised ImageNet-1K fine-tuning.

cal remote-sensing transfer, and SAR transfer. The BigEarthNet-S2/S1 CNN pair provides the cleanest domain contrast: architecture, dataset, and supervised land-cover task are shared, while sensing modality differs. The ViT comparison is best-available rather than method-matched because SatDINO uses DINO [1] and SARMAE uses masked autoencoding [6]. The ImageNet pair shares its final dataset and classification supervision, but not its full history: the ViT uses supervised AugReg [10], whereas ConvNeXt-V2 uses FCMAE followed by supervised ImageNet-1K fine-tuning.

Detector. xView3 labels are points and its metric matches predictions by distance, not box IoU, so we use a point-native heatmap head in the style of CenterNet / Objects as Points [13]: each vessel is a Gaussian blob on a stride-4 map, decoded by peak-finding with distance-based non-maximum suppression. Every arm receives the same fixed 3-channel input [VH, VV, VH-VV] in dB, so input handling is never a confound between arms.

Fine-tuning and comparability. All arms fine-tune end-to-end through one code path with shared AdamW, augmentation, and seed 0. Training runs for up to 50 epochs with early stopping on dev-split F1; the decision threshold is selected on dev and frozen before test. Continuous-integration guard tests make comparability a checked invariant rather than a claim: a backbone-parity test asserts identical within-track parameter counts and matching head-adaptor geometry, a checkpoint-load test asserts that each pre-trained backbone loads exact non-random weights, and a scorer-immutability test freezes the evaluation code (Figure 1).

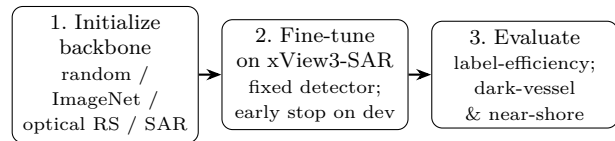


Figure 1. The three stages, run identically for every arm; only the backbone initialization varies within a track.

3. Data and Evaluation

We use locally acquired xView3 DIU-format Sentinel-1 GRD archives with dual polarization VH/VV. The frozen seed-0 split contains 150 study scenes: 111 train, 23 dev, and 16 test. Training fractions are nested and selected at the scene level (10/25/50/100%), preventing chip leakage between splits. A separate set of exactly 50 human-verified scenes forms eval_final; it is evaluated once, after the grid and operating thresholds are frozen, with no subsequent tuning. Dark-vessel labels occur only in this verified set, so dark-vessel recall is not reported on dev or test. LS-SSDD is not consumed by the active study. The controlled grid reports eight seed-0 detection-F1 curves as point estimates, with near-shore F1 and once-only final dark-vessel recall. YOLO26 [7] and LocateAnything-3B are the only external references and remain off the curves.

4. Timeline and Feasibility

All six pretrained initializations are downloaded, so local training is limited to 32 seed-0 xView3 detector fine-tunes; with the two completed external references, the full plan contains 34 experiments. Six core 10%-fraction cells and both references are complete, leaving 26 core cells. Completed RTX 5070 Ti timings project roughly 19–22 days of continuous single-GPU execu-

tion for those cells, excluding data transfer and final evaluation. The historical V100 forecast is retired, and measured P100 throughput failed the project's compute tripwire; the remaining recipe-identical one-GPU jobs are assigned to the RTX 5070 Ti. We target the WACV 2027 Round-2 deadline (paper registration August 21, 2026; submission August 28, 2026, AoE).

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